Solving Polynomial Systems of using Linear Elimination

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Abstract

There is a problem shared among many fields of science of solving polynomial systems of equations. The required solution is often a space of solutions, rather than a single solution. Applications include QFT, robotics, cryptography, optimization, chemistry, electrical engineering, and more. [we need other examples... best is to have concrete examples from history. If not then maybe say generally that in many areas of science and engineering a parametric solution can surprise us and help where our creativity is limited].

This paper presents a scalable pipeline for finding a space of solutions (of full dimension) to polynomial systems of equations. We start from numerical solutions, transitioning to symbolic solutions, and finally parametrize the solution space via structured exploration. We then also discuss other complementary methods and discuss what open questions still remain. We demonstrate these methods on concrete examples from the field of QFT.

Introduction

Solving polynomial systems of equations is a fundamental problem in mathematics and computer science, with applications in fields such as robotics, cryptography, optimization, QFT, chemistry, electrical engineering, economics and more. In QFT, polynomial systems arise in the context of generilizing and unifying theories (e.g. quantum gravity). In chemistry, for example, polynomial systems appear in the study and design of reaction chains. In robotics, for example, polynomial systems are used to model kinematics and motion planning.

In most of these use cases, the required solution is often a space of solutions, rather than a single solution. Specifically, in QFT the proposed theory must follow

gauge invariance, which leads to a space of solutions, and of these solutions one may want to find the one represents the simplest theory. In chemistry, reaction chains often have multiple valid pathways, leading to a solution space. In robotics, the solution space may represent all possible configurations of a robotic arm that satisfy certain constraints. one would want to work with the entire solution space to find optimal paths and configurations. This type of problem is universal across many fields of science and engineering. It calls for a comprehensive approach that can efficiently find and represent the solution space of polynomial systems.

Classic methods for solving polynomial systems of equations include Groebner bases [], resultants [], and homotopy continuation []. However, these methods can be computationally expensive and may not scale well with the size of the system. Other approaches, such as gradient [], and SMT solvers [] are focused on finding specific solutions rather than exploring the entire solution space.

In this paper, classify/group/review/discuss specially made approaches for finding the general ... in efficient ways relevant for. We also demonstrate the methods on concrete examples from QFT. Among those methods, we propose a pipeline that leverages numerical methods to find initial solutions, then refines these solutions into symbolic forms, and finally explores a space of solutions. This approach aims to combine the efficiency of numerical methods, and some insights we may have at how the symbolic solutions may look like, to provide a comprehensive understanding of a solution space of polynomial systems. We also present and analyze a complementary examples from QFT, showcasing the effectiveness of our approach in practical scenarios. We finally outline the open questions and future directions [more concrete] in this field.

Mathematical Background

Let $\mathbb{K}[\mathbf{x}]$ denote the polynomial ring in variables $\mathbf{x} = (x_1, \ldots, x_n)$ over a field $\mathbb{K}$, typically $\mathbb{F}_q$, $\mathbb{R}$ or $\mathbb{C}$. A polynomial system is defined by a map $F : \mathbb{K}^n \to \mathbb{K}^m$ with components $f_1, \ldots, f_m \in \mathbb{K}[\mathbf{x}]$. The locus of simultaneous solutions constitutes the *affine variety* $V(F) = \{\mathbf{a} \in \mathbb{K}^n \mid F(\mathbf{a}) = \mathbf{0}\}$.

In this paper, we will mostly work over the rational, real or complex numbers.

Linear elimination

even in the case of very large non linear polynomial systems of equations, it is possible to eliminate variables in a linear fashion, which can significantly reduce the size of

the system and the number of variables.

Algorithm 1 Linear elimination by substitution

Require: Polynomial system $P = \{p_1, \ldots, p_k\} \subseteq \mathbb{K}[x_1, \ldots, x_m]$

Ensure: Reduced system Q where no $q \in Q$ is linear in any remaining variable; and a substitution map Σ to recover eliminated variables

$Q \leftarrow P$
$V \leftarrow \{x_1, \ldots, x_m\}$
$\Sigma \leftarrow \emptyset$
$L \leftarrow \{(x', p') \mid x' \in V,\ p' \in Q,\ p' \text{ is linear in } x'\}$
while $L \neq \emptyset$ **do**
    Select any pair $(x, p) \in L$
    Solve $p = 0$ for x to obtain $x \leftarrow f(\hat{x})$
    $\Sigma \leftarrow \Sigma \cup \{x \mapsto f(\hat{x})\}$
    Substitute $x \leftarrow f(\hat{x})$ in Q
    $V \leftarrow V \setminus \{x\}$
    Simplify the system Q ▷ Optional, but recommended
    clear Q of zero equations
    $L \leftarrow \{(x', p') \mid x' \in V,\ p' \in Q,\ p' \text{ is linear in } x'\}$
end while
if $Q = \emptyset$ and $\Sigma \neq \emptyset$ **then**
    Validate: back-substitute Σ into each $p \in P$; each resulting polynomial must be identically zero (probabilistic zero test in the implementation)
end if
return (Q, Σ)

we have the freedom to choose which linear variable to eliminate at each step. good heuristics to choose which variable to eliminate include:

- **Lowest degree in the other equations:** eliminate the variable that appears with the lowest degree in the other equations first.
- **Lowest degree after substitution:** eliminate the variable that will lead to the lowest degree in the other equations after substitution.
- **Greatest or least number of occurrences:** eliminate the variable that appears in the greatest or least number of equations first.

It is also possible to use DFS or BFS to search on the order of elimination.

Validation of the solution

After obtaining a solution, we can validate it by substituting the solution back into the original system and checking if it satisfies all the equations.

To check whether the solution is of full dimension at certain points, we can use the following theorem:

Theorem 1 (Gatilov et al. [1]) *Given a continuously differentiable system $F \in C^1$ (possibly polynomial) with a solution point p, the dimension of the solution space in the neighborhood of p is $n - rank(J_F(p))$, where J_F is the Jacobian matrix of F at point p.*

System dataset

For linear elimination we need to be able to do the following operations efficiently

- is polynomial linear in a variable
- is polynomial zero
- linearly eliminate a variable from a polynomial

To prevent an exponential blowup in the size of the equations, we store the current polynomial system in unexpanded form. Each polynomial expression is represented as a DAG, so common subexpressions are shared and elimination steps do not force an immediate expansion of all dependent equations. The algorithm therefore works directly with graph-represented polynomials.

Once the system is stored in this form, linearity and vanishing can no longer be decided from the top-level syntax alone. Instead, the required operations are phrased in terms of evaluation of the polynomial represented by the DAG.

The techniques outlined here can be immediately extended to rational functions.

Evaluating a DAG-represented expression

Let G be a DAG representing a polynomial expression, and let α be an assignment of values to the variables occurring in G. To evaluate the represented expression, we process the nodes of G in topological order, so that every child of a node is evaluated before the node itself.

Algorithm 2 Evaluating a polynomial expression represented as a DAG

Require: DAG G with root r, and an assignment α of values to the variables of G
Ensure: The value of the polynomial represented by G at α

for each node v of G in topological order **do**
  if v is a constant node with value c **then**
    $\mathrm{val}(v) \leftarrow c$
  else if v is a variable node labeled by x **then**
    $\mathrm{val}(v) \leftarrow \alpha(x)$
  else if v is an addition node with children $u_1, \ldots, u_k$ **then**
    $\mathrm{val}(v) \leftarrow \mathrm{val}(u_1) + \cdots + \mathrm{val}(u_k)$
  else if v is a multiplication node with children $u_1, \ldots, u_k$ **then**
    $\mathrm{val}(v) \leftarrow \mathrm{val}(u_1) \cdots \mathrm{val}(u_k)$
  else if v is a negation node with child u **then**
    $\mathrm{val}(v) \leftarrow -\mathrm{val}(u)$
  end if
end for
return $\mathrm{val}(r)$

Because the expression is represented as a DAG rather than a tree, a shared subexpression appears as a single node and is evaluated only once. The values computed at intermediate nodes can therefore be reused wherever that subexpression occurs, which makes repeated evaluations efficient and supports the tests for linearity and vanishing described above.

Sampling points

For quick evaluation of polynomials, we will sample the random rational numbers $\frac{p}{q}$ such that $|p|, |q| < n$.

Lemma 1 *The number of destinct rational numbers we sample from is* $4(\sum_{i=1}^{n} \phi(i)) - 1$*, where* $\phi(i)$ *is the Euler's totient function. Asymptotically, this is* $\frac{12}{\pi^2} n^2 + O(n \log n)$*.*

Proof 1 *Consider the positive rational numbers* $\frac{p}{q}$ *in lowest terms (i.e.,* $\gcd(p, q) = 1$*) such that* $1 \leq p, q \leq n$*. We can partition these into two sets: those where* $p \leq q$ *and those where* $p \geq q$*. For a fixed denominator* q*, the number of numerators* $p \leq q$ *coprime to* q *is exactly Euler's totient function* $\phi(q)$*. Summing over all possible denominators* $1 \leq q \leq n$*, the number of fractions with* $p \leq q$ *is* $\sum_{i=1}^{n} \phi(i)$*. By*

symmetry, the number of fractions with $q \leq p$ is also $\sum_{i=1}^{n} \phi(i)$. The only fraction counted twice is $\frac{1}{1}$, so the total number of strictly positive rational numbers is $2\sum_{i=1}^{n} \phi(i) - 1$.

Since we can independently choose the sign of the rational number, there is an equal number of strictly negative rational numbers. Adding the single case of 0 (represented as $\frac{0}{1}$), the total number of distinct rational numbers is:

$$2\left(2\sum_{i=1}^{n}\phi(i) - 1\right) + 1 = 4\sum_{i=1}^{n}\phi(i) - 1.$$

For the asymptotic bound, it is a well-known result in number theory [2] that the summatory totient function grows as:

$$\sum_{i=1}^{n}\phi(i) = \frac{3}{\pi^2}n^2 + O(n\log n).$$

Multiplying this by 4 yields the final asymptotic bound of $\frac{12}{\pi^2}n^2 + O(n\log n)$.

If a polynomial has m free variables, then the number of points we can sample from is $|S| = (4(\sum_{i=1}^{n}\phi(i)) - 1)^m$.

Is polynomial zero

For the operation "is polynomial zero" we will use Schwartz-Zippel Lemma. We evaluate the represented expression at c sampled points of its remaining free variables. If every sampled value is zero, we treat the equation as zero. The likelihood of a polynomial $p \neq 0$ evaluating to zero at c sampled points is $(\frac{d}{|S|})^c$, where d is the degree of the polynomial [3, 4].

Is polynomial linear in a variable

The operation "is polynomial linear in a variable" can be viewed as determining if a polynomial is zero. let p be a polynomial and without the loss of generality let x_1 be a variable, we will run the zero test on the following polynomial:

$$\begin{aligned} q(u, v, w, x_2, \ldots, x_n) = &(p(w, x_2, \ldots, x_n) - p(u, x_2, \ldots, x_n))(v - u) \\ &- (p(v, x_2, \ldots, x_n) - p(u, x_2, \ldots, x_n))(w - u) \end{aligned}$$

Lemma 2 *The polynomial q is zero if and only if p is linear in x_1.*

Proof 2 *If p is linear in x_1, then there are a, b constant in x_1 such that $p(x_1, x_2, \ldots, x_n) = ax_1 + b$. Then*

$$\begin{aligned} q(u, v, w, x_2, \ldots, x_n) &= (aw + b - (au + b))(v - u) - (av + b - (au + b))(w - u) \\ &= a(w - u)(v - u) - a(v - u)(w - u) \\ &= 0. \end{aligned}$$

Conversely, assume q is zero, then for all $u, v, w, x_2, \ldots, x_n$, we have

$$\begin{aligned} 0 &= q(u, v, w, x_2, \ldots, x_n) \\ &= (p(w, x_2, \ldots, x_n) - p(u, x_2, \ldots, x_n))(v - u) \\ &\quad - (p(v, x_2, \ldots, x_n) - p(u, x_2, \ldots, x_n))(w - u) \end{aligned}$$

Thus we have

$$\frac{p(w, x_2, \ldots, x_n) - p(u, x_2, \ldots, x_n)}{w - u} = \frac{p(v, x_2, \ldots, x_n) - p(u, x_2, \ldots, x_n)}{v - u}$$

We get that there are a, b constant in x_1 such that $p(x_1, x_2, \ldots, x_n) = ax_1 + b$, and we get that p is linear in x_1.

Linearly eliminate a variable from a polynomial

For the operation "linearly eliminate a variable from a polynomial", suppose that the represented equation is linear in x_1, so it can be written as

$$ax_1 + b = 0,$$

where x is the variable to be eliminated. Since the equation is linear in x, both a and b may depend on the other variables but not on x. These coefficients can be recovered by evaluation for $s, e, m \in \mathbb{F}$, $a = \frac{p(e, x_2, \ldots, x_n) - p(s, x_2, \ldots, x_n)}{e - s}$ and $b = p(m, x_2, \ldots, x_n) - am$. Usally we will choose $e = 1$ and $s = m = 0$, but we can choose other values to avoid division by zero.

The substitution of x will be divided into cases

- $a, b = 0$ we discard the equation
- $a = 0$ and $b \neq 0$ This is an illegal step, because the equation cannot be solved for x
- $a \neq 0$ we discard the equation and substitute $x \mapsto -\frac{b}{a}$ without expanding the expression.

Complexity Analysis

This section analyzes the cost of the operations introduced in the previous section.

Cost of the primitive operations

The basic subroutine is evaluation of a represented expression.

Lemma 3 *Evaluating the expression represented by G at a fixed assignment takes $O(|G|)$ time.*

Proof 3 *The evaluation algorithm processes the nodes of G in topological order. Each node is visited once, and each visit performs only constant-time work beyond combining the already computed values of its children.*

Cost of the tests from the substitution-map representation

Lemma 4 *Let p be a represented polynomial with DAG G.*

1. *Testing whether $p \equiv 0$ by evaluating it at c sampled points takes $O(|G|)$ time.*
2. *Testing whether p is linear in a chosen variable x_1 takes $O(|G|)$ time.*
3. *If p is linear in x_1, recovering the represented coefficients a and b in*

$$p(x_1, \ldots, x_n) = ax_1 + b$$

by the identities $b = p(0, x_2, \ldots, x_n)$ and $a = p(1, x_2, \ldots, x_n) - p(0, x_2, \ldots, x_n)$ takes $O(|G|)$ time.

Proof 4 *For the zero test, we perform c evaluations of G, one at each sampled point thus the total cost is $O(|G| \cdot c) = O(|G|)$.*

For the linearity test, we perform a zero test for a polynomial with a constant number of additional nodes, thus the total cost is $O(|G|)$.

Coefficient recovery involves a constant number of copies and evaluations of G, thus the total cost is $O(|G|)$.

One elimination step and overall growth

Let $N(Q, \Sigma)$ denote the total number of DAG nodes appearing in the equations in Q and the substitution map Σ. We assume that arithmetic in the base field and memoized node lookup both take constant time.

Lemma 5 *One elimination step starting from (Q_i, Σ_i) has cost $O(N_i)$, where $N_i = N(Q_i, \Sigma_i)$.*

Proof 5 *Let (p, x) be the chosen pair. By the previous lemma, testing whether p is linear in x and recovering the coefficients needed to solve for x cost $O(|p|)$. Forming $p(0)$ and $p(1)$ and building the represented substitution*

$$x \mapsto -\frac{b}{a}$$

add only $O(|p|)$ work, since they duplicate or evaluate p a constant number of times. Removing the active equations that have become zero costs $O(N_i)$, because each zero test is linear in the size of its DAG and the total size of the active equations is at most N_i. Since $|p| \le N_i$, the whole step costs $O(N_i)$.

Moreover, the substitution is stored symbolically rather than expanded through all active equations, so one successful elimination step adds only a constant number of represented expressions derived from p. Hence there is a constant K such that

$$N_{i+1} \le N_i + K|p| \le (1 + K)N_i.$$

Theorem 2 *The total running time of the elimination phase is*

$$O\left(\sum_{i=0}^{d-1} N_i\right) = O(N_0 c_0^d).$$

Proof 6 *First note that for $c_0 = 1 + K$ we have $N_{i+1} \le c_0 N_i$ for every successful elimination step; iterating gives $N_d \le c_0^d N_0$. Since each step costs $O(N_i)$, the total runtime is $O(\sum_{i=0}^{d-1} N_i) = O(N_0 c_0^d)$.*

In summary, all basic operations are linear in the size of the current represented state. Each successful elimination step adds only a constant amount of new represented structure, because substitutions are kept symbolic instead of being expanded throughout the system. As a result, the represented state grows by at most a constant factor per step, and the total runtime is exponential in the elimination depth.

Handling systems without linearities

the polynomial order we will use to interreduce, and make polynomials monic, should maximize the number of monomials with that are linear in one of any of the variables. thus we should use the grevlex order.

Experiments

Experimental Setup

The implementation of our method was written in Rust and is available at `https://github.com/IlayMenahem/soothing`. We ran our experiments on m4 pro cpu with 14 cores and 24GB of memory.

Results

Conclusion and Future Work

References

[1] Stepan Yu Gatilov. Using low-rank approximation of the jacobian matrix in the newton–raphson method to solve certain singular equations. *Journal of Computational and Applied Mathematics*, 272:8–24, 2014.

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[3] Jacob T Schwartz. Fast probabilistic algorithms for verification of polynomial identities. *Journal of the ACM (JACM)*, 27(4):701–717, 1980.

[4] Richard Zippel. Probabilistic algorithms for sparse polynomials. In *Symbolic and Algebraic Computation: EUROSAM'79, An International Symposium on Symbolic and Algebraic Computation Marseille, France, June 1979 Proceedings*, pages 216–226. Springer, 1979.

Appendix

Simplifying expressions

To simplify an expression, we can use the following local simplifications. Some are always semantics-preserving, while others are valid only under additional side conditions.

Always safe local simplifications.

- $x_1 + \ldots + 0 + \ldots + x_n \mapsto x_1 + \ldots + x_n$
- $x_1 + \ldots + x + \ldots + (-x) + \ldots + x_n \mapsto x_1 + \ldots + x_n$
- $-(-x) \mapsto x$
- $x_1 \cdot \ldots \cdot 1 \cdot \ldots \cdot x_n \mapsto x_1 \cdot \ldots \cdot x_n$
- $x_1 \cdot \ldots \cdot 0 \cdot \ldots \cdot x_n \mapsto 0$
- $((-1)^{n_1} \cdot x_1) \cdot \ldots \cdot ((-1)^{n_k} \cdot x_k) \mapsto (-1)^{n_1 + \ldots + n_k} \cdot (x_1 \cdot \ldots \cdot x_k)$ where n_i are integers
- $\frac{x}{1} \mapsto x$
- $\frac{x}{y} \cdot z \mapsto \frac{x \cdot z}{y}$
- $\frac{-x}{y} \mapsto -\frac{x}{y}$
- $\frac{x}{-y} \mapsto -\frac{x}{y}$
- $\frac{-x}{-y} \mapsto \frac{x}{y}$
- $\frac{x_1}{y_1} \cdot \ldots \cdot \frac{x_n}{y_n} \mapsto \frac{x_1 \cdot \ldots \cdot x_n}{y_1 \cdot \ldots \cdot y_n}$
- $\frac{\frac{x}{y}}{\frac{z}{w}} \mapsto \frac{x \cdot w}{y \cdot z}$
- $x^1 \mapsto x$
- $1^n \mapsto 1$
- constant folding for addition, multiplication, division, and powers whenever the result is defined, if the result is undefined give an error.

- flatten nested additions and multiplications: $x_1+\ldots+(y_1+\ldots+y_m)+\ldots+x_n \mapsto x_1+\ldots+y_1+\ldots+y_m+\ldots+x_n$
- combine terms: $y_1 \cdot x + \ldots + y_k \cdot x \mapsto (y_1 + \ldots + y_k) \cdot x$

Local simplifications requiring side conditions.

- $\frac{x}{x} \mapsto 1$ if $x \neq 0$
- $\frac{0}{y} \mapsto 0$ if $y \neq 0$
- $\frac{x \cdot y}{x \cdot z} \mapsto \frac{y}{z}$ if $x \neq 0$ and $z \neq 0$
- $\frac{x_1 \cdot \ldots \cdot x_i \cdot \ldots \cdot x_n}{y_1 \cdot \ldots \cdot x_i \cdot \ldots \cdot y_m} \mapsto \frac{x_1 \cdot \ldots \cdot x_{i-1} \cdot x_{i+1} \cdot \ldots \cdot x_n}{y_1 \cdot \ldots \cdot y_{i-1} \cdot y_{i+1} \cdot \ldots \cdot y_m}$ if $x_i \neq 0$ and $y_i \neq 0$
- $x^0 \mapsto 1$ if $x \neq 0$, or if 0^0 is defined by convention
- $0^n \mapsto 0$ for positive integers n

Local simplifications algorithm. To apply all the local simplifications, we use a fixed-point algorithm over the expression DAG. First we find all the local simplifications that can be applied to the DAG. Then while there are still local simplifications that can be applied, we apply them and look at the changed nodes to see if they introduce new possible simplifications.

If there are multiple possible simplifications, we apply them in the following order of precedence:

1. **Flattening:** For associative operations (addition and multiplication), flatten nested identical operations. This exposes all operands to the subsequent steps.
2. **Combining like terms:** Group constants together (partial constant folding among siblings) and combine affine terms (e.g., $a \cdot x + b \cdot x \mapsto (a + b) \cdot x$).
3. **Annihilators and Identities:** Apply rules involving 0 and 1 (e.g., $x \cdot 0 \mapsto 0$, $x + 0 \mapsto x$, $x^1 \mapsto x$).
4. **Sign and Fraction Normalization:** Push negations down or pull them up consistently, and simplify fractions (e.g., $-(-x) \mapsto x$, $\frac{-x}{-y} \mapsto \frac{x}{y}$).
5. **Constant folding:** If all operands are constants, immediately compute the result.

DFS for linear elimination order

In this setting, DFS means that we choose one available linear elimination, apply it, and continue recursively on the reduced system until no further linear elimination is possible. Only then do we backtrack and try a different elimination choice from an earlier step.

Algorithm 3 DFS search for a linear elimination order

Require: Polynomial system P
Ensure: A reduced system Q together with a substitution map Σ

function DFSLinearElimination(P, Σ)
  $L \leftarrow \{(p, x) \mid p \in P,\ p \text{ is linear in } x\}$
  if $L = \emptyset$ or $P = \emptyset$ **then**
    return (P, Σ)
  end if
  $B \leftarrow (P, \Sigma)$
  for all $(p, x) \in L$ **do**
    solve $p = 0$ for x to obtain $x \leftarrow f(\hat{x})$
    $P' \leftarrow P$ with $x \leftarrow f(\hat{x})$ substituted in every equation
    remove zero equations from P'
    $\Sigma' \leftarrow \Sigma \cup \{x \mapsto f(\hat{x})\}$
    $R \leftarrow$ DFSLinearElimination(P', Σ')
    if $R = (\emptyset, \Sigma'')$ for some Σ'' **then**
      return R
    end if
    if R is better than B **then**
      $B \leftarrow R$
    end if
  end for
  return B
end function

Here "better" means that the resulting system has fewer remaining equations, and if the number of remaining equations is the same, then we prefer the system with more free variables.